When are Imprecise Bandits Computationally Tractable?

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1. Setting

Let $A \subseteq \mathbb{R}^{d_A}$ be the arm set and let $D \subseteq \mathbb{R}^{d_D}$ be the outcome set. We assume that A and D are Euclidean balls of radii R_A and R_D (not necessarily centred at the origin). An element $x \in A$ is called an **arm**, while an element $y \in D$ is a possible sampled outcome. The reward is affine in the arm and outcome:

$$r(x, y) := a^T x + b^T y + c.$$

Let $Z \subseteq \mathbb{R}^{d_Z}$ denote the hypothesis set, and let $z^\star \in Z$ be the unknown true hypothesis. For every $z \in Z$ and $x \in A$, define

$$K_z(x) := \{y \in D : C_z y + B_z x + d_z = 0\}. \tag{1}$$

Here B_z, C_z, and d_z of are z-dependent matrices and vectors of appropriate dimensions.

The true feasible outcome set is $K^\star(x) := K_{z^\star}(x)$.

We assume throughout that $K^\star(x)$ is nonempty for every $x \in A$.

Let $N := \ker C_{z^\star}$.

Lemma 1. *There exists an affine map $f_z : A \to \mathbb{R}^{d_D}$ and a linear subspace N_z independent of x such that*

$$f_z(x) + N_z = \{y \in \mathbb{R}^{d_D} : C_z y + B_z x + d_z = 0\}, \quad x \in A.$$

Consequently, there exists an f and N such that

$$K^\star(x) = (f(x) + N) \cap D, \quad x \in A.$$

Proof. Fix x and let $y_1, y_2 \in K_z(x)$. Then $C_z(y_1 - y_2) = 0$, which means $y_1 - y_2$ must lie in a subspace determined by z. We can pick some $y \in K_z(x)$ arbitrarily and set $f_z(x) = y$. This is clearly an affine mapping. ■

Note that as the proof demonstrates, f_z need not be unique. Indeed, for each z there can be multiple f_z's that satisfy the requirement of the lemma. However, each z determines a unique N_z.

We also assume a uniform non-tangency condition: there is a constant $S \in (0, 1]$ such that, for every $x \in A$ and every $p \in \mathbb{R}^{d_D}$ satisfying $C_{z^\star} p + B_{z^\star} x + d_{z^\star} = 0$ with $p \notin D$,

$$\operatorname{dist}(p, D) \geq S \operatorname{dist}(p, K^\star(x)).$$

For a Euclidean ball, this says that the affine solution spaces defined by the true constraints remain uniformly bounded away from tangency to D. This is the transversality condition used in [1].

The interaction proceeds for rounds $t = 1, ..., T$. Let $\mathcal{F}_{t-1}$ denote the history before round t. The learner chooses an arm $x_t \in A$. Nature may then choose, adaptively as a function of the history and x_t, any distribution supported on D whose conditional mean

$$m_t := E[y_t \mid \mathcal{F}_{t-1}, x_t]$$

belongs to $K^\star(x_t)$. An outcome y_t is sampled from this distribution, and the learner receives reward $r(x_t, y_t)$.

The robust value of an arm and the optimal robust value are

$$v^\star(x) := \min_{y \in K^\star(x)} r(x, y), \quad V^\star := \max_{x \in A} v^\star(x).$$

The regret over horizon T is

$$R_T := TV^\star - \sum_{t=1}^{T} r(x_t, y_t).$$

Finally, let

$$C_r := \max_{x \in A, y \in D} r(x, y) - \min_{x \in A, y \in D} r(x, y)$$

denote the reward range.

Theorem 1 (efficient $T^{\frac{2}{3}}$ learning). *Suppose that the known problem data in the setting above are rational. For every known horizon T, there is a policy whose arithmetic and weak-optimization running time is polynomial in T and the input bit length and which, for every true hypothesis satisfying the uniform non-tangency condition and every compatible adaptive nature policy, satisfies*

$$E[R_T] \le \tilde{O}\Big(P(d_A, d_D, R_A, R_D, \|b\|_2, C_r, S^{-1})T^{\frac{2}{3}}\Big),$$

where P is a fixed polynomial. Apart from the uniform non-tangency condition on the true feasible sets, no geometric property of Z is used.

2. Warmup: Noiseless Setting

In this section, playing an arm x reveals an exact feasible response $y \in K^\star(x)$; there is no sampling noise. However, there still is the Knightian uncertainty corresponding to the adversary picking an arbitrary outcome inside $K^\star(x)$.

Lemma 2. *Suppose that, for every arm $x \in A$, we have identified a nonempty set $K'(x) \subseteq K^\star(x)$, and suppose that whenever the learner plays x, nature chooses an outcome in $K'(x)$. Define*

$$v'(x) \coloneqq \min_{y \in K'(x)} r(x, y),$$

and choose

$$x' \in \operatorname{argmax}_{x \in A} v'(x).$$

Then playing x' on every round incurs nonpositive regret with respect to the original robust benchmark, i.e. $R_T \le 0$.

Lemma 3. *There exist $d_A + 1$ arms $x^{(0)}, ..., x^{(d_A)} \in A$ such that, from any possible sequence of responses $y^{(i)} \in K^\star\big(x^{(i)}\big)$, we can compute an affine map $\hat{f}$ such that*

$$K^\star(x) = \Big(\hat{f}(x) + N\Big) \cap D, \quad x \in A.$$

Proof. Let f be any affine map supplied by Lemma 1. Let $x^{(0)}, ..., x^{(d_A)}$ be any affine basis of A. Play them in sequence and let $y^{(0)}, ..., y^{(d_A)}$ be the corresponding outcomes chosen by the adversary. Let $\hat{f}$ be the unique affine interpolator for which $\hat{f}\big(x^{(i)}\big) = y^{(i)}$ for all i. Since $y^{(i)} \in K^\star\big(x^{(i)}\big)$, we have that $\hat{f}\big(x^{(i)}\big) + N = f\big(x^{(i)}\big) + N$ for all i. Since the $x^{(i)}$ form an affine basis, $\hat{f}(x) + N = f(x) + N$ for all $x \in A$. ■

Algorithm 1: Noiseless imprecise bandit.

1. Play the $d_A + 1$ arms from Lemma 3 and use their responses to construct $\hat{f}$.
2. Initialize $N_{d_A+2} \coloneqq \{0\}$.
3. For each round $t = d_A + 2, ..., T$:
    - For every $x \in A$, define the provisional feasible set
    $$K'_t(x) \coloneqq \Big(\hat{f}(x) + N_t\Big) \cap D.$$
    - If $K'_t(x) = \varnothing$ for some $x \in A$, choose any such arm as x_t. Otherwise, choose
    $$x_t \in \operatorname{argmax}_{x \in A} \min_{y \in K'_t(x)} r(x, y).$$
    - Play x_t and observe y_t.
    - If $y_t - \hat{f}(x_t) \notin N_t$, set
    $$N_{t+1} \coloneqq \operatorname{span}\Big(N_t \cup \Big\{y_t - \hat{f}(x_t)\Big\}\Big).$$
    Otherwise, set $N_{t+1} \coloneqq N_t$.

Every N_t maintained by Algorithm 1 is a subspace of N. Whenever it is updated, its dimension increases by one. Moreover, if a provisional feasible set is empty, playing an arm with an empty provisional set necessarily triggers such an update. There can therefore be at most $\dim(N) \le d_D$ update rounds.

On every remaining round, all the provisional feasible sets are nonempty and the observed response belongs to $K'_t(x_t)$. Hence Lemma 2 shows that the reward on that round is at least the original robust benchmark. Charging at most a constant regret to each of the $d_A + 1$ initial rounds and to each update round gives $R_T \le O(d_A + d_D)$.

2.1. Computational Tractability

All the steps in Algorithm 1 can be carried out in polynomial time. First, affine interpolation can be done via Gaussian elimination.

The subspace N_t can be stored through an orthonormal basis. Given the new residual

$$h_t := y_t - \hat{f}(x_t),$$

project h_t onto $N_t^{\perp}$. If the projection is zero, the span does not change. Otherwise, normalize the projection and append it to the stored basis. This is an incremental Gram–Schmidt and takes polynomial time.

It remains to compute the arm in the planning step.

Lemma 4. *Fix a round t. Given rational descriptions of $A, D, \hat{f}$, and N_t, one can compute, in polynomially many arithmetic operations, an arm $x_{\mathrm{emp}} \in A$ such that, if $K_t'(x) = \varnothing$ for some $x \in A$, then $K_t'(x_{\mathrm{emp}}) = \varnothing$.*

Proof. Clearly, for a given x, the set $K_t'(x)$ is empty if and only if the distance between c_D and $\hat{f}(x) + N_t$ is bigger than R_D. Thus to decide if there exists an $x \in A$ for which $K_t'(x)$ is empty, we need to find the arm that maximizes the distance between c_D and $\hat{f}(x) + N_t$. To see why this can be done in polynomial time, let $q_t(x)$ be the point in $\hat{f}(x) + N_t$ that is closest to c_D. Note that $q_t(x)$ can be written as an affine function.

$$q_t(x) := c_D + (I - P_t)\big(\hat{f}(x) - c_D\big).$$

Here P_t is the projection on N_t.

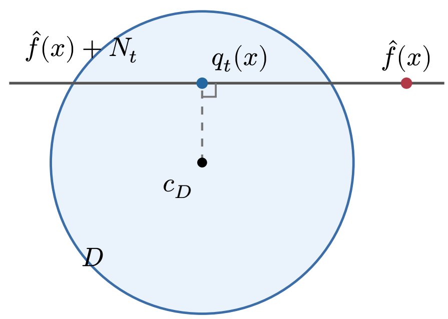

Figure 1: Two-dimensional outcome geometry when N_t is one-dimensional. The dashed segment is orthogonal to $\hat{f}(x) + N_t$, so its endpoint is $q_t(x)$, the point on that line closest to c_D.

Now we need to compute

$$x_{\mathrm{emp}} \in \mathrm{argmax}_{x \in A} \, \|q_t(x) - c_D\|_2^2.$$

This is a quadratic optimization problem over the Euclidean ball A, which is polynomial-time solvable using the result in [2]. ■

Lemma 5. *Fix a round t and suppose that $K_t'(x)$ is nonempty for every $x \in A$. Let the rational problem data have total bit length L. For every $\varepsilon \in (0, 1)$, one can compute an arm $x_\varepsilon \in A$ in time* $\mathrm{poly}\big(L, \log(\frac{1}{\varepsilon})\big)$ *such that*

$$\min_{y \in K_t'(x_\varepsilon)} r(x_\varepsilon, y) \geq \max_{x \in A} \min_{y \in K_t'(x)} r(x, y) - \varepsilon.$$

Proof. Compute the orthogonal projector P_t onto N_t as in the proof of Lemma 4, and define

$$q_t(x) := c_D + (I - P_t)\big(\hat{f}(x) - c_D\big).$$

Since every provisional set is nonempty, Pythagoras' theorem gives

$$K_t'(x) = q_t(x) + \left\{ u \in N_t : \|u\|_2 \leq \sqrt{R_D^2 - \|q_t(x) - c_D\|_2^2} \right\}.$$

Put $\beta_t := \|P_t b\|_2$. For a fixed arm x, minimizing the linear reward over the displayed ball moves in direction $-P_t b$, and hence

$$v_t'(x) := \min_{y \in K_t'(x)} r(x, y) = a^T x + b^T q_t(x) + c - \beta_t \sqrt{R_D^2 - \|q_t(x) - c_D\|_2^2}.$$

Introduce a scalar $s \geq 0$. Maximizing v_t' is equivalent to the QCQP

$$\max_{x,s} a^T x + b^T q_t(x) + c - \beta_t s$$

subject to

$$\|x - c_A\|_2^2 \le R_A^2, \quad s^2 + \|q_t(x) - c_D\|_2^2 = R_D^2, \quad s \ge 0,$$

where c_A is the centre of A. Represent the equality by two quadratic inequalities. If $R_A, R_D > 0$, add the redundant constraint

$$\frac{\|x - c_A\|_2^2}{R_A^2} + \frac{s^2}{R_D^2} \le 2.$$

This is an ellipsoid in the joint variable (x, s). Thus the QCQP has a fixed number of quadratic constraints, one of which is ellipsoidal. Bienstock's theorem [3] returns a weakly feasible, additive δ-optimal solution in time polynomial in L and $\log\left(\frac{1}{\delta}\right)$. The cases $R_A = 0$ or $R_D = 0$ reduce immediately to a lower-dimensional problem.

To obtain a feasible arm, project the x-component of the weak solution onto A, call it x_δ, and set

$$s_\delta := \sqrt{R_D^2 - \|q_t(x_\delta) - c_D\|_2^2}.$$

The square root is real by the assumption of the lemma. The projection moves x by $O\left(\sqrt{\delta}\right)$, and the violated quadratic equality implies that s changes by at most $O\left(\delta^{\frac{1}{4}}\right)$. Consequently the linear objective changes by at most $C\delta^{\frac{1}{4}}$, where $C \ge 1$ is a data-dependent constant whose bit length is polynomial in L. Choose $\delta \le \left(\frac{\varepsilon}{2C}\right)^4$ and approximate β_t to the same working precision. The resulting arm is feasible and its robust value is within ε of the optimum. Since $\log\left(\frac{1}{\delta}\right) = \mathrm{poly}(L) + O\left(\log\left(\frac{1}{\varepsilon}\right)\right)$, the running time is polynomial. ■

Taking $\varepsilon = T^{-2}$ adds at most T^{-1} to total regret. Since the provisional model changes only when N_t grows, the planning problem is solved at most $d_D + 1$ times. With finite-precision data, the strict comparison in Lemma 4 uses the usual gray-zone convention: if the maximum squared distance is within $\varepsilon_{\mathrm{emp}}$ of R_D^2, enlarge the squared outcome radius by $\varepsilon_{\mathrm{emp}}$. This changes the radius by only $O\left(\sqrt{\varepsilon_{\mathrm{emp}}}\right)$ and is absorbed into the noisy tolerances below.

3. Noisy Setting

In this section, constants hidden by O and $\tilde{O}$ may depend on the fixed problem parameters. In addition, $\tilde{O}$ suppresses logarithmic factors in T and $\frac{1}{\delta}$.

In the noisy setting, estimating the true N is tricky. To reduce the level of noise, we can work with with average outcomes over blocks of length n instead of each individual outcome. Suppose m_i is the conditional mean corresponding to the outcome y_i. Then over a block of size n, we can bound $\left\|\frac{1}{n}\sum y_i - \frac{1}{n}\sum m_i\right\| \le O\left(n^{-\frac{1}{2}}\right)$ with high probability. Due to the convexity of D, $\frac{1}{n}\sum m_i$ also lies in D, and therefore, the "residuals" calculated using $\overline{y} = \frac{1}{n}\sum y_i$ should be close to the true residuals. However, it is still risky to compute the subspace containing this residual and committing to it for the future steps. A small error in the direction of the calculated subspace can lead to large errors further out. Thus to get better control over the estimation errors, we approximate the subspace N with an ellipsoid. We start by computing an approximate $\hat{f}$ using an affine basis, and assume that N is a unit ball of radius λ where $\lambda \le O\left(n^{-\frac{1}{2}}\right)$. Then if a residual arrives that lies outside the current ellipsoid, we expand the ellipsoid in a specific way along the direction of the residual.

Lemma 6. *Fix $\delta \in (0,1)$ and $1 \le n \le T$. There exist $d_A + 1$ affinely independent anchor arms such that, after playing each arm for n rounds and interpolating their empirical average outcomes, the resulting affine map $\hat{f}$ satisfies the following with probability at least $1 - \delta$. There is an affine map f and a linear subspace N such that $K^\star(x) = (f(x) + N) \cap D$ and*

$$\sup_{x \in A} \left\|\hat{f}(x) - f(x)\right\|_2 \le \tilde{O}\left(n^{-\frac{1}{2}}\right).$$

Proof. Let $x^{(0)}, ..., x^{(d_A)}$ be any affine basis of A. Play each arm for n rounds. Let $\overline{y}^{(i)}$ be the average observed outcome for arm i, and let $\overline{m}^{(i)}$ be the corresponding average conditional mean. Since $K^\star\left(x^{(i)}\right)$ is convex, $\overline{m}^{(i)} \in K^\star\left(x^{(i)}\right)$. Let $\hat{f}$ and f be the unique affine interpolators satisfying $\hat{f}\left(x^{(i)}\right) = \overline{y}^{(i)}$ and $f\left(x^{(i)}\right) = \overline{m}^{(i)}$ for every i. By Lemma 3, $K^\star(x) = (f(x) + N) \cap D$ for every $x \in A$.

Within each block, $y_t - m_t$ is a martingale-difference sequence. Since D is bounded, the Azuma–Hoeffding inequality for bounded martingale differences (see [4], [5]), together with a union bound over the finitely many anchors and outcome coordinates, implies that, with probability at least $1-\delta$,

$$\max_{0 \le i \le d_A} \left\| \overline{y}^{(i)} - \overline{m}^{(i)} \right\|_2 \le \tilde{O}\left(n^{-\frac{1}{2}}\right).$$

Now consider an arbitrary $x \in A$, and let $\alpha_0, ..., \alpha_{d_A}$ be its affine coordinates with respect to the chosen basis. Since A is compact, there is a constant L, independent of x, such that

$$\sum_{i=0}^{d_A} |\alpha_i| \le L.$$

Since $\hat{f}$ and f are affine and agree with the corresponding values at the anchors,

$$\left\| \hat{f}(x) - f(x) \right\|_2 \le \sum_{i=0}^{d_A} |\alpha_i| \left\| \overline{y}^{(i)} - \overline{m}^{(i)} \right\|_2 \le L\tilde{O}\left(n^{-\frac{1}{2}}\right).$$

Taking the supremum over $x \in A$ proves the result. ■

In the noisy setting, taking the span of the observed residuals is unstable: even a small amount of noise can introduce a spurious direction. We therefore replace the subspace N_t by a learned residual ellipsoid.

Fix $\delta \in (0,1)$ and $1 \le n \le T$, and let $\hat{f}$ and f be as in Lemma 6. One can choose $\lambda = \tilde{O}\left(n^{-\frac{1}{2}}\right)$ with $\lambda \ge n^{-\frac{1}{2}}$ so that, conditional on the anchor conclusion, the conclusions of the following two lemmas hold simultaneously throughout the horizon with probability at least $1-\delta$.

Algorithm 2: Noisy imprecise bandit (n, λ).

Parameters: block length $n \in \mathbb{N}$ and ridge parameter $\lambda > 0$.

1. Choose the $d_A + 1$ anchor arms from Lemma 6. Play each anchor for n consecutive rounds, compute its empirical average $\overline{y}^{(i)}$, and construct the affine interpolant $\hat{f}$. If the horizon ends during this phase, stop.
2. Initialize $M := 0$ and let $\hat{G}_0$ be the matrix with no columns.
3. At the beginning of each subsequent block, define
$$V_M := \lambda^2 I_{d_D} + \hat{G}_M \hat{G}_M^T \in \mathbb{R}^{d_D \times d_D}, \quad \mathcal{E}_M := \left\{u \in \mathbb{R}^{d_D} : u^T V_M^{-1} u \le 1\right\}.$$
For every arm $x \in A$, define
$$\hat{K}_M(x) := \left(\hat{f}(x) + \mathcal{E}_M\right) \cap D.$$
4. If $\hat{K}_M(x) = \varnothing$ for some $x \in A$, choose any such arm. Otherwise, choose
$$x \in \operatorname{argmax}_{x' \in A} \min_{y \in \hat{K}_M(x')} r(x', y).$$
5. Play the chosen arm for n rounds. If fewer than n rounds remain, play until the horizon and stop without updating. Otherwise, let $\overline{y}$ be the block-average outcome and set
$$\hat{g} := \overline{y} - \hat{f}(x).$$
6. If $\hat{g} \notin \mathcal{E}_M$, declare the block informative, set $\hat{G}_{M+1} := \left[\hat{G}_M, \hat{g}\right]$, and then increase M by one. Otherwise leave $\hat{G}_M$ and M unchanged. Return to Step 3 and continue until time T.

Lemma 7. *At most $O(\log T)$ blocks of Algorithm 2 are informative.*

Proof. Enumerate the stored residuals as $\hat{g}_1, ..., \hat{g}_M$. After j of them have been stored,

$$V_j := \lambda^2 I_{d_D} + \sum_{i=1}^{j} \hat{g}_i \hat{g}_i^T.$$

Every block average lies in D. The anchor averages also lie in D, and affine interpolation from the fixed anchors has uniformly bounded coefficients on the compact set A. Hence $\left\|\hat{g}_j\right\|_2 = O(1)$ for every block.

When $\hat{g}_j$ is stored, the block is informative, so

$$\hat{g}_j^T V_{j-1}^{-1} \hat{g}_j > 1.$$

The matrix determinant lemma [6] therefore gives

$$\det\left(V_j\right) = \det\left(V_{j-1}\right)\left(1 + \hat{g}_j^T V_{j-1}^{-1} \hat{g}_j\right) > 2 \det\left(V_{j-1}\right).$$

Since $V_0 = \lambda^2 I_{d_D}$, iteration yields

$$\det(V_M) > 2^M \lambda^{2d_D}.$$

On the other hand, $M \leq \frac{T}{n}$, and therefore

$$\operatorname{tr}(V_M) = d_D \lambda^2 + \sum_{j=1}^{M} \left\|\hat{g}_j\right\|_2^2 \leq d_D \lambda^2 + O\left(\frac{T}{n}\right).$$

Applying the arithmetic–geometric mean inequality to the eigenvalues of V_M gives

$$\det(V_M) \leq \left(\lambda^2 + O\left(\frac{T}{n}\right)\right)^{d_D}.$$

Combining the two determinant bounds,

$$M \leq d_D \log_2\left(1 + O\left(\frac{T}{n\lambda^2}\right)\right) = O(\log T),$$

where the last equality uses $\lambda \geq n^{-\frac{1}{2}}$. ■

Lemma 8. *Let $\hat{f}$ be the empirical affine interpolant constructed during the anchor phase of Algorithm 2, and let f be the affine interpolant of the corresponding average conditional means, as in Lemma 6. For every current ellipsoid $\mathcal{E}_M$, every $x \in A$, and every $u \in \mathcal{E}_M$,*

$$\operatorname{dist}\left(\hat{f}(x) + u, f(x) + N\right) \leq \tilde{O}\left(n^{-\frac{1}{2}}\right).$$

Proof. We prove by induction on M.

We first show that this bound holds for the initial matrix $V_0 = \lambda^2 I_{d_D}$, and then show that it remains true after every update from V_M to V_{M+1}.

For the initial matrix $V_0 = \lambda^2 I_{d_D}$, the ellipsoid $\mathcal{E}_0$ is the Euclidean ball of radius λ. By Lemma 6, there exists some $\beta_n \leq \tilde{O}\left(n^{-\frac{1}{2}}\right)$ such that $\left\|\hat{f}(x) - f(x)\right\|_2 \leq \beta_n$ for every $x \in A$. Therefore, by the triangle inequality, for every $x \in A$ and $u \in \mathcal{E}_0$,

$$\operatorname{dist}\left(\hat{f}(x) + u, f(x) + N\right) \leq \left\|\hat{f}(x) - f(x)\right\|_2 + \|u\|_2 \leq \beta_n + \lambda \leq \tilde{O}\left(n^{-\frac{1}{2}}\right).$$

Thus the claimed bound holds when $M = 0$.

Geometrically, $\hat{f}(x) + \mathcal{E}_0$ is a ball of radius λ centred at $\hat{f}(x)$, while $f(x) + N$ is the affine space that we want this ball to remain close to. Each informative update adds a new generating direction $\hat{g}$ to the ellipsoid. The component of $\hat{g}$ lying in N only moves the ellipsoid parallel to $f(x) + N$ and therefore does not increase the distance from that affine space. Only the component orthogonal to N matters. Thus, if $\hat{g}$ is within η_n of N, one update can increase the distance by at most η_n. An uninformative block does not change the ellipsoid at all.

To make this precise, consider a complete block $\mathcal{B}$ at arm x and let

$$\overline{m} := \frac{1}{n} \sum_{t \in \mathcal{B}} m_t.$$

Since $\overline{m} \in K^\star(x)$, the vector $g := \overline{m} - f(x)$ belongs to N. The empirical residual is $\hat{g} := \overline{y} - \hat{f}(x)$. Conditionally on the history at the beginning of any adaptively selected block, the vectors $y_t - m_t$ remain martingale differences. The same concentration argument as in Lemma 6, followed by a union bound over the at most $\frac{T}{n}$ complete blocks, therefore gives, simultaneously for every block,

$$\|\overline{y} - \overline{m}\|_2 \leq \tilde{O}\left(n^{-\frac{1}{2}}\right).$$

Combining this with the uniform anchor-interpolation bound gives

$$\|\hat{g} - g\|_2 \leq \tilde{O}\left(n^{-\frac{1}{2}}\right).$$

We may therefore choose $\eta_n = \tilde{O}\left(n^{-\frac{1}{2}}\right)$ so that every stored residual satisfies $\operatorname{dist}(\hat{g}, N) \leq \eta_n$. We prove the following more explicit form of the induction claim:

$$\operatorname{dist}\left(\hat{f}(x) + u, f(x) + N\right) \leq \beta_n + \lambda + M\eta_n, \quad x \in A, u \in \mathcal{E}_M.$$

The base case above establishes this claim when $M = 0$.

Now consider an update from V_M to V_{M+1}. Suppose that the bound holds for M and that the informative block producing the update has residual $\hat{g}$. Let $Q_M := \left[\hat{G}_M, \lambda I_{d_D}\right]$. Since $Q_M Q_M^T = V_M$ and Q_M has full row rank,

$$\mathcal{E}_M = \{Q_M\theta : \|\theta\|_2 \le 1\}.$$

After storing $\hat{g}$, every $u \in \mathcal{E}_{M+1}$ can therefore be written as

$$u = Q_M\theta + a\hat{g}, \quad \|\theta\|_2^2 + a^2 \le 1.$$

Set $u_0 := Q_M\theta$. Then $u_0 \in \mathcal{E}_M$, and since N is a linear subspace, for every $x \in A$,

$$\operatorname{dist}\Big(\hat{f}(x) + u, f(x) + N\Big) \le \operatorname{dist}\Big(\hat{f}(x) + u_0, f(x) + N\Big) + |a| \operatorname{dist}(\hat{g}, N) \le \beta_n + \lambda + (M+1)\eta_n.$$

This proves the induction claim. By Lemma 7, $M = O(\log T)$, and therefore

$$\beta_n + \lambda + M\eta_n = \tilde{O}\Big(n^{-\frac{1}{2}}\Big).$$

■

Lemma 9. *Under the uniform non-tangency condition, for every $x \in A$ and every $y \in \mathbb{R}^{d_D}$,*

$$\operatorname{dist}(y, K^\star(x)) \le \big(1 + S^{-1}\big) \operatorname{dist}(y, f(x) + N) + S^{-1} \operatorname{dist}(y, D).$$

In particular, if $y \in D$, then

$$\operatorname{dist}(y, K^\star(x)) \le \big(1 + S^{-1}\big) \operatorname{dist}(y, f(x) + N).$$

Proof. Let p be the Euclidean projection of y onto $f(x) + N$. If $p \in D$, then $p \in K^\star(x)$ and the result is immediate. Otherwise, the uniform non-tangency condition and the triangle inequality give

$$\operatorname{dist}(p, K^\star(x)) \le S^{-1} \operatorname{dist}(p, D) \le S^{-1}(\|y - p\|_2 + \operatorname{dist}(y, D)).$$

Since $\|y - p\|_2 = \operatorname{dist}(y, f(x) + N)$, one more application of the triangle inequality proves the first claim. The second follows by setting $\operatorname{dist}(y, D) = 0$. ■

Lemma 10. *Suppose that the conclusions of Lemma 6 and Lemma 8 hold. Let $\mathcal{B}$ be a complete n-round block that is declared uninformative, and let x be the arm played in that block. Then its realized average regret satisfies*

$$\frac{1}{n}\sum_{t\in\mathcal{B}} (V^\star - r(x, y_t)) \le \tilde{O}\Big(n^{-\frac{1}{2}}\Big).$$

Proof. Let M be the state at the start of the block and let

$$\overline{y} := \frac{1}{n}\sum_{t\in\mathcal{B}} y_t.$$

Since the block is uninformative, $\hat{g} := \overline{y} - \hat{f}(x)$ belongs to $\mathcal{E}_M$.

Since D is convex, $\overline{y} \in D$. Therefore,

$$\overline{y} \in \Big(\hat{f}(x) + \mathcal{E}_M\Big) \cap D = \hat{K}_M(x).$$

In particular, this block could not have been selected by the empty-set branch of Step 4, so its arm was selected by the planning rule. Define

$$\hat{v}_M(x') := \min_{y\in\hat{K}_M(x')} r(x', y).$$

For every $x' \in A$ and $y \in \hat{K}_M(x')$, Lemma 8 gives

$$\operatorname{dist}(y, f(x') + N) \le \tilde{O}\Big(n^{-\frac{1}{2}}\Big).$$

Since $y \in D$, applying Lemma 9 gives

$$\operatorname{dist}(y, K^\star(x')) \le \tilde{O}\Big(n^{-\frac{1}{2}}\Big).$$

Since the reward is Lipschitz in its outcome argument,

$$\hat{v}_M(x') \ge v^\star(x') - \tilde{O}\Big(n^{-\frac{1}{2}}\Big).$$

Let $x^\star$ maximize $v^\star$. The planning rule and the fact that $\overline{y} \in \hat{K}_M(x)$ now give

$$r(x, \overline{y}) \ge \hat{v}_M(x) \ge \hat{v}_M(x^\star) \ge V^\star - \tilde{O}\Big(n^{-\frac{1}{2}}\Big).$$

Finally, the reward is affine in the outcome and the arm is fixed throughout the block, so

$$r(x, \overline{y}) = \frac{1}{n}\sum_{t\in\mathcal{B}} r(x, y_t).$$

This proves the average-regret bound. ■

Taking $\delta = (T+1)^{-2}$, $n = \Big\lceil T^{\frac{2}{3}} \Big\rceil$, and λ chosen as above now gives the claimed regret rate. The anchor phase, the $O(\log T)$ informative blocks from Lemma 7, and the final partial block contribute $\tilde{O}(n)$ regret.

By Lemma 10, all remaining rounds contribute $\tilde{O}\left(Tn^{-\frac{1}{2}}\right)$. Hence, when the anchor conclusion and the joint ridge conclusion both hold,

$$R_T \le \tilde{O}\left(n + Tn^{-\frac{1}{2}}\right) = \tilde{O}\left(T^{\frac{2}{3}}\right).$$

These conclusions fail with probability at most $2(T+1)^{-2}$, and $R_T \le C_r T$ always, so taking expectations only adds $O\left(\frac{C_r}{T}\right)$.

3.1. Computational Tractability

The statistical argument above presents Algorithm 2 using exact emptiness tests and exact maximization over the single provisional set $\hat{K}_M(x)$. We now explain how to implement these operations with weak finite-precision optimization. The additional sets introduced in this section provide numerical slack only; they do not change the statistical idea.

The matrix updates, leverage calculations, and affine interpolation are standard polynomial-time linear algebra. Moreover, $V_M - \lambda^2 I_{d_D}$ is positive semidefinite, while $\lambda \ge T^{-\frac{1}{2}}$ and the stored residuals have bounded norm. Hence all inversions can be carried out to the required inverse-polynomial precision in polynomial time. For this implementation, one may choose the centre c_A of A and the arms $c_A + R_A e_i$, $i = 1, ..., d_A$, as the affine basis. Their interpolation coefficients and the resulting affine map are computed by Gaussian elimination and matrix multiplication, with polynomially controlled bit complexity. This conditioning detail is irrelevant to the dependence on n and was therefore omitted from the statistical argument.

3.1.1. Empty-Fiber Certification

Fix $\rho := \lambda$, let c_D be the centre of D, and define

$$D_1 := \left\{y : \|y - c_D\|_2 \le R_D + \rho\right\}, \quad D_2 := \left\{y : \|y - c_D\|_2 \le R_D + 2\rho\right\},$$

$$\hat{K}_M^{\text{in}}(x) := \left(\hat{f}(x) + 2\mathcal{E}_M\right) \cap D_1, \quad \hat{K}_M^{\text{out}}(x) := \left(\hat{f}(x) + 3\mathcal{E}_M\right) \cap D_2.$$

The implementation tests the inner sets for emptiness and plans against the outer sets. The exact set used in the statistical proof satisfies

$$\hat{K}_M(x) \subseteq \hat{K}_M^{\text{in}}(x) \subseteq \hat{K}_M^{\text{out}}(x).$$

The constants 2 and 3 are not important. Their role, together with the two expanded outcome balls, is to create a strict margin between the set containing an uninformative empirical outcome and the set used for planning.

Write $\hat{f}(x) = Fx + f_0$. For a fixed arm, the distance between $\hat{f}(x) + 2\mathcal{E}_M$ and D_1 is

$$d_M(x) := \max_{\|u\|_2 \le 1} \left(u^T\left(\hat{f}(x) - c_D\right) - 2\sqrt{u^T V_M u} - (R_D + \rho)\|u\|_2\right).$$

This is the support-function formula for the distance between two closed convex sets. In particular, $d_M(x) > 0$ exactly when $\hat{K}_M^{\text{in}}(x) = \varnothing$.

After introducing nonnegative variables s_0, s_1 satisfying

$$u^T V_M u \le s_0^2, \quad \|u\|_2^2 \le s_1^2,$$

maximizing $d_M(x)$ jointly over $x \in A$ and $\|u\|_2 \le 1$ becomes a QCQP with a fixed number of quadratic constraints. The only coupling between the arm and dual variables is the bilinear expression $u^T F x$. We impose the explicit bounds $\|u\|_2 \le 1$, $0 \le s_1 \le 1$, and $0 \le s_0 \le \sqrt{\lambda_{\max}(V_M)}$, together with the ball constraint on x, and combine them into a redundant positive-definite enclosing ellipsoid. The weak fixed-constraint result in Section A then applies.

A weakly feasible output is corrected before it is used as a certificate: project its x and u components onto their respective balls, reset s_0, s_1 to the corresponding exact norms, and directly evaluate the support-function objective. Since all variables are polynomially bounded, using sufficiently smaller internal accuracy makes the change in objective at most ζ. The corrected point is feasible, so a positive value is a genuine certificate of emptiness.

Exact emptiness classification is unnecessary. Let $\zeta < \frac{\min(\rho,\lambda)}{10}$ and solve the preceding maximization to additive accuracy ζ. If an arm has certified distance greater than 2ζ, its inner set is genuinely empty. Otherwise every inner pair of constituent sets is at distance at most 3ζ. The corresponding outer intersection then contains a Euclidean ball of radius at least

$$\min(\lambda, \rho - 3\zeta).$$

Indeed, the positive semidefiniteness of $V_M - \lambda^2 I_{d_D}$ implies that $\mathcal{E}_M$ contains the Euclidean ball of radius λ. The gap from $2\mathcal{E}_M$ to $3\mathcal{E}_M$ therefore supplies the ellipsoid margin, while the gap from D_1 to D_2 supplies the outcome-ball margin.

3.1.2. Robust Planning

The preceding strict-feasibility margin also makes robust planning tractable. Fenchel–Rockafellar duality gives the following dual representation (see [7]):

$$\min_{y \in \widehat{K}_M^{\mathrm{out}}(x)} r(x, y) = a^T x + c + b^T c_D + \max_{u \in \mathbb{R}^{d_D}} \left(u^T \left(\hat{f}(x) - c_D \right) - 3\sqrt{u^T V_M u} - (R_D + 2\rho)\|b - u\|_2 \right).$$

The interior ball bounds the norm of an optimal dual vector by a polynomial in the input size, T, and $\min(\lambda, \rho - 3\zeta)^{-1}$. Introducing epigraph variables for the two square roots therefore turns the joint maximization over x and u into a bounded QCQP with a fixed number of quadratic constraints. Appendix A gives an additive-ζ_v optimal arm in polynomial time. As in the emptiness problem, a weakly feasible solution is projected onto the arm ball and its epigraph variables are conservatively increased. The resulting loss is absorbed into ζ_v.

3.1.3. Finite-Precision Comparisons

The remaining comparisons use the same kind of gray zone. Let $\tilde{y}$ denote the rounded block average used by the implementation, and choose the rounding precision τ so that $\|\tilde{y} - \overline{y}\|_2 \le \tau \le \rho$. Define the residual used in the leverage test to be $\hat{g} := \tilde{y} - \hat{f}(x)$. Since $\overline{y} \in D$, the expanded ball D_1 absorbs the rounding error and $\tilde{y} \in D_1$.

Let $\ell := \hat{g}^T V_M^{-1} \hat{g}$ and compute an approximation $\tilde{\ell}$ satisfying $\left|\tilde{\ell} - \ell\right| \le \frac{\gamma}{2}$. Append the residual only if $\tilde{\ell} - \frac{\gamma}{2} > 1$. Such an update has $\ell > 1$ and therefore retains the determinant-doubling argument in Lemma 7. Otherwise $\ell \le 1 + \gamma$, so, for $\gamma \le 1$,

$$\hat{g} \in \sqrt{1+\gamma}\,\mathcal{E}_M \subseteq 2\mathcal{E}_M.$$

Finally, the proof of Lemma 8 is unchanged when $\mathcal{E}_M$ is multiplied by any fixed constant. Thus points in the outer ellipsoid remain $\tilde{O}\left(n^{-\frac{1}{2}}\right)$-close to $f(x) + N$, while points in D_2 are at distance at most 2ρ from D. Applying Lemma 9 shows that every point in an outer fiber is $\tilde{O}\left(n^{-\frac{1}{2}}\right)$-close to the corresponding true feasible set.

It remains to connect this numerical wrapper to the statistical proof. If a complete block is not appended, its rounded residual lies in $2\mathcal{E}_M$, and therefore

$$\tilde{y} \in \hat{K}_M^{\mathrm{in}}(x) \subseteq \hat{K}_M^{\mathrm{out}}(x).$$

Consequently, a complete block selected using a certified empty inner fiber must be appended; otherwise the displayed membership would contradict emptiness. Every complete block that is not appended is therefore a planning block. For such a block, define

$$\hat{v}_M^{\mathrm{out}}(x') := \min_{y \in \widehat{K}_M^{\mathrm{out}}(x')} r(x', y).$$

The preceding outer-fiber bound and the Lipschitz property of the reward give, uniformly in x',

$$\hat{v}_M^{\mathrm{out}}(x') \ge v^\star(x') - \tilde{O}\left(n^{-\frac{1}{2}}\right).$$

If the weak planner returns an additive-ζ_v maximizer and $x^\star$ maximizes $v^\star$, then

$$r(x, \overline{y}) \ge r(x, \tilde{y}) - \|b\|_2 \tau \ge \hat{v}_M^{\mathrm{out}}(x) - \|b\|_2 \tau \ge \hat{v}_M^{\mathrm{out}}(x^\star) - \zeta_v - \|b\|_2 \tau \ge V^\star - \tilde{O}\left(n^{-\frac{1}{2}}\right) - \zeta_v - \|b\|_2 \tau.$$

Affineness again identifies $r(x, \overline{y})$ with the realized average reward in the block. Thus certified empty-fiber blocks and leverage updates are charged to the same $O(\log T)$ informative blocks as before, while every other complete block has average regret at most $\tilde{O}\left(n^{-\frac{1}{2}}\right) + \zeta_v + \|b\|_2 \tau$.

Taking $\rho = \lambda = \tilde{O}\left(n^{-\frac{1}{2}}\right)$ and choosing, for example, $\zeta = (T+1)^{-8}$, $\gamma = (T+1)^{-8}$, $\zeta_v = (T+1)^{-3}$, and $\tau = (T+1)^{-12}$ makes every numerical error negligible or absorbs it into the existing $\tilde{O}\left(n^{-\frac{1}{2}}\right)$ bound. These choices require only $O(\log T)$ additional precision bits, and the planning error contributes at most $T\zeta_v$ to regret. Hence the weak bit-model implementation remains polynomial-time and has the same regret rate.

4. NP-hardness of IUCB

Given rational descriptions of A, D, Z, the affine reward r, and the maps defining $K_z(x)$, let

$$v_z(x) := \min_{y \in K_z(x)} r(x, y).$$

The **exact IUCB optimism problem** asks for the value

$$V_{\mathrm{IUCB}} := \max_{z \in Z, x \in A} v_z(x).$$

This is the optimization performed on the first round of IUCB, when its confidence set is all of Z. The associated search problem asks for a globally optimal first arm

$$x^\star \in \mathrm{argmax}_{x \in A} \max_{z \in Z} v_z(x).$$

Theorem 2. *Even when A, D, and Z are Euclidean balls and planning for every fixed hypothesis is polynomial-time, computing the exact first optimistic value or a globally optimal first arm of IUCB is NP-hard.*

Proof. We reduce from the exact tensor spectral-norm problem. Given a rational order-three tensor $\mathcal{T} \in \mathbb{Q}^{m \times d \times d_A}$, viewed as a bilinear map $\mathcal{T} : \mathbb{R}^d \times \mathbb{R}^{d_A} \to \mathbb{R}^m$, this problem asks to compute

$$\|\mathcal{T}\|_\sigma := \max_{\|u\|_2 \le 1, \|x\|_2 \le 1} \|\mathcal{T}(u, x)\|_2.$$

Computing this value is NP-hard. From $\mathcal{T}$, construct an IUCB instance as follows. Put

$$L := 1 + \sum_{i,j,k} |T_{ijk}|, \quad \alpha := \frac{1}{2L}, \quad A := \{x : \|x\|_2 \le 1\}.$$

Write $z = (z_0, u)$ and $y = (w, s)$, and take

$$Z := \left\{ (z_0, u) : \left(z_0 - \frac{5}{3}\right)^2 + \|u\|_2^2 \le 1 \right\}, \quad D := \{(w, s) : \|w\|_2^2 + s^2 \le 1\}.$$

For $z = (z_0, u)$, define the data in Equation 1 and the reward by

$$B_z x := -\alpha \mathcal{T}(u, x), \quad C_z(w, s) := z_0 w, \quad d_z := 0, \quad r(x, (w, s)) := \frac{1}{2}(1 + s).$$

These are instances of the setting above, and

$$K_z(x) = \left\{ (w, s) \in D : w = \frac{\alpha}{z_0} \mathcal{T}(u, x) \right\}.$$

The definition of Z gives $z_0 \ge \frac{2}{3}$ and

$$\frac{\|u\|_2}{z_0} \le \frac{3}{4},$$

because $1 - \left(z_0 - \frac{5}{3}\right)^2 - 9\frac{z_0^2}{16} = -\frac{(15z_0 - 16)^2}{144} \le 0$. Equality holds at $z_0 = \frac{16}{15}$ and $\|u\|_2 = \frac{4}{5}$ in every direction. Also $\|\mathcal{T}(u, x)\|_2 \le L\|u\|_2\|x\|_2$, so the fixed w in $K_z(x)$ has norm at most $3\alpha\frac{L}{4} = \frac{3}{8}$. Thus every $K_z(x)$ is nonempty, and the uniform non-tangency condition holds with $S \ge \frac{\sqrt{55}}{8}$. For this instance, nature chooses $s = -\sqrt{1 - \|w\|_2^2}$ when evaluating $v_z(x)$, whence

$$v_z(x) = h\left(\frac{\alpha}{z_0}\|\mathcal{T}(u, x)\|_2\right), \quad h(t) := \frac{1}{2}\left(1 - \sqrt{1 - t^2}\right).$$

For fixed z, a top right singular vector of $M_u x := \mathcal{T}(u, x)$ maximizes $v_z(x)$, so fixed-hypothesis planning is polynomial-time. IUCB instead optimizes jointly over Z and A. Since h is strictly increasing and the ratio above attains $\frac{3}{4}$ in every direction,

$$V_{\mathrm{IUCB}} = h\left(3\frac{\alpha}{4}\|\mathcal{T}\|_\sigma\right),$$

Since

$$\|\mathcal{T}\|_\sigma = \frac{4}{3\alpha}\sqrt{1 - (1 - 2V_{\mathrm{IUCB}})^2},$$

exact IUCB optimism would solve the tensor spectral-norm problem. If IUCB returns only a globally optimal arm $x^\star$, one SVD of $N_x u := \mathcal{T}(u, x^\star)$ computes

$$\max_{z \in Z} v_z(x^\star) = h\left(3\frac{\alpha}{4}\sigma_{\max}(N_x)\right) = V_{\mathrm{IUCB}},$$

so the same inverse recovers $\|\mathcal{T}\|_\sigma$. Producing a globally optimal first arm is therefore NP-hard as well. ■

A. Quadratically constrained quadratic programming

For symmetric matrices $Q_i \in \mathbb{R}^{d \times d}$, a **quadratically constrained quadratic program** (QCQP) in $z \in \mathbb{R}^d$ has the form

$$\min_{z \in \mathbb{R}^d} g_0(z) \quad \text{subject to} \quad g_i(z) \le 0, \quad i = 1, ..., m,$$

where $g_i(z) := z^T Q_i z + 2p_i^T z + r_i$; for complexity statements, all entries are rational. A quadratic equality $h(z) = 0$ is exactly the pair $h(z) \le 0$ and $-h(z) \le 0$ [3].

QCQP describes algebraic form, whereas convex programming describes geometry. The displayed minimization problem is a convex program when every Q_i, $i = 0, ..., m$, is positive semidefinite and all equalities are affine [7]. Without those restrictions it can be nonconvex: general QCQP is NP-hard even when only the objective is quadratic and its Hessian has one negative eigenvalue [8].

The polynomial-time conclusion for our nonconvex instance has two steps.

1. First, fix the number m of constraints. Bienstock's theorem states that, for rational quadratics $g_0, ..., g_m$, if at least one constraint $g_i(z) \le 0$ has a positive-definite quadratic part, then for every $\varepsilon \in (0, 1)$ an algorithm runs in time $\mathrm{poly}\big(L, \log\big(\frac{1}{\varepsilon}\big)\big)$, where L is the input bit length, and either proves infeasibility or returns $\hat{z}$ such that
$$g_i(\hat{z}) \le \varepsilon, \quad i = 1, ..., m,$$
and $g_0(\hat{z}) \le g_0(z) + \varepsilon$ for every feasible z [3]. The underlying weak-feasibility step reduces to a fixed number of homogeneous quadratic equations on a sphere, handled by Barvinok's method; binary search then gives the objective guarantee [3], [9].
2. Second, our planning problem is a QCQP with a fixed number of constraints, one of which is the strictly convex joint-ellipsoid constraint when $R_A, R_D > 0$. Thus it satisfies Bienstock's hypotheses. Applying the theorem to the negative objective computes the planning maximum to weak additive accuracy ε in time $\mathrm{poly}\big(L, \log\big(\frac{1}{\varepsilon}\big)\big)$. If either radius is zero, the problem first reduces to a lower-dimensional instance.

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